

MOL3000/MOL3100 – Applied Bioinformatics – Computer Lab

Searching protein databases with PSI-Blast

(A note about the formatting: `Courier` font indicates text you should type in, `Arial` font indicates links, buttons, menu items etc.)

Finding the protein sequence.

Go to `http://www.ncbi.nlm.nih.gov/`. In the **Search** pull down menu (top of page) select **Protein** and in the text box type `AIJ06171`. Click the **Search** button.

In the result window look at the sequence record. Are you able to find anything about the function of the protein in the description?

Getting the sequence in correct (fasta) format.

Click on the **Fasta** link (below the entry title). The display will now change to show the sequence in Fasta format. *Keep this window.*

Running the PSI-Blast search.

In a new window go to `http://www.ncbi.nlm.nih.gov/`. Click on **Blast** in the menu for Popular Resources (right side), then click on the **Protein BLAST** link under **Web BLAST**.

In the sequence window from the first step select the sequence, including the info line (starting with `>`), and copy it. Go to the Blast window and paste it into the **Query sequence** text box. Make sure that **Protein Data Bank proteins (pdb)** is selected as **Database**, and **PSI-BLAST (Position-Specific Iterated BLAST)** as **Algorithm**.

(The choice of database is partly a trick to make the exercise slightly more interesting. The input sequence is very similar to a large number of recently published genomic sequences. We see the effect of iterative searching better if we leave these out and focus on sequences with known 3D structure.)

At the bottom of the page, find **Algorithm parameters**, and click on that. Find **General Parameters** and set **Expect threshold** to `10`. This will return more non-significant hits, more similar to the lecture. Find **PSI/PHI/DELTA BLAST** and change **PSI-BLAST Threshold** from `0.005` to `0.05`. This will make the search more sensitive, but also increase the risk that false positive hits are included.

Now click on the **BLAST** button. The window will change. You get a **Request ID**, and you can use that to retrieve the search result later. Now you are going to just wait for the result.

After a while, the window will be updated, and this will be repeated until you get the search result. This may take some time, depending on how many users are running Blast jobs. Do not resubmit your job just because you are getting impatient, as that will only increase the waiting time for everyone, including you!

Now take a look at the result, trying to understand the different parts of the output. Look at the plot of hits, the list of hits and the alignment of hits. Do the hits give you any indication of what type of protein this might be? Do you believe that this indication may be correct?

Now start a second iteration by clicking on the **Run** button next to **Run PSI-Blast iteration 2**. Wait until the window has been updated with the new result.

Now look carefully at the result window and try to understand how it has changed. Are there many new hits? Do these new hits strengthen or weaken your initial hypothesis about the function of this protein?

Now run a third iteration, using the same procedure as for the second iteration. Evaluate your search result again, as above.

Looking at the search result.

Look in particular at the alignment output. Test various links and try to find out what type of information they can give you.

Look at **Taxonomy**. Study the output and try to understand the different parts. There are 3 different reports (Linage, Organism and Taxonomy), the last report is probably easiest to understand. How many hits do you have from each of the main groups of organisms (Archaea, Bacteria, Eukaryota)? Are all of the hits listed in this table significant? Hint: Look at the Organism report and the E values listed there.

Doing another search.

Now do a similar search for the sequence BAA97652. Again, use the Protein Data Bank proteins (pdb) in Blast to limit the number of hits (use the Database pulldown menu on the Blast page), but set PSI-BLAST Threshold back to 0.005. After the third iteration search the list of results for entries annotated as Trypsin (e.g. by using the search option in the browser). Use the first or one of the first entries and go to the corresponding alignment. You can do this by clicking on the name in the list of Sequences producing significant alignments. Compare the alignment to the alignment in the paper by Jones and Swindells (TiBS 27(3), 161-164 (2002)) shown below. Are the essential residues for trypsins as indicated in their alignment (red numbers) also conserved in your alignment? That is, are you able to confirm that ExTox (your query sequence) may have trypsin-like activity by showing that the positions corresponding to essential residues for trypsins (indicated with red arrows) are conserved in your query sequence? (Hint: your trypsin will be different from the trypsin shown below, and therefore the numbering will be different. But you can try to find the relevant sequence fragments in ExTox below (e.g. NRHIA), find the same fragment in your query sequence in the alignment, and see if the central residue (here H) is conserved in the alignment.)

ExTox	:	GQTSATGV	LIGKNTV	LTNRHIA	KFANGD	PSKVS	FRPSIN	TTDDNG	NTETPY	GEYEV	KEILQ				
		G	G	L+	+N	V++	H	K	+V	G+	+	++			
Trypsin:		GYHFCG	SLVNEN	WVVSAA	H	CYKSR	---	VEVRL	GEHNI	KVTEG	SEQFI	SSSRVIRHPNY			
							57								
ExTox	:	EPFGAG	VDLAL	IRLKP	DQNGV	SLGDK	ISP	AKIGTS	NDLKD	GDKLE	LIGYP	FDHKVNQMHR			
		+	D+	LI+L		+L	+	P	+	TS	G	+	G+	+	++
Trypsin:		SSYNID	NDIM	LKLS	---	KPATL	NTYV	QPVAL	PTS-	CAPAG	TMCTV	SGW--	GNTMS	STAD	
				102											
ExTox	:	SE-----	IELTT-----	LSRGL	RYYG	F	TVP-----	GNSG	SGIF	NSNGEL	VGI				
		S		L+		++	+	G+		G+SG	+	NGEL	G+		
Trypsin:		SNKLQ	CLNIP	ILSYSD	CNN	SYPGM	ITNAM	FCAGY	LEGGK	DSCQ	GD	SGGP	VVC-	NGELQ	GV
											195				
ExTox	:	HSSKV	SHLDRE	HQIN	YGV--	GIGNYV									
		S		+	+	Y		+++							
Trypsin:		VSWG	YGCAE	PGNPG	VYAKV	CIF	NDWL								

T/BS

TiBS

Done!

You do *not* need to submit a report on this computer lab.